

## PROGRAM B: SSTF DISK SCHEDULING

### C PROGRAM

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    int req[20], visited[20] = {0};
```

```
    int n, head, i, count = 0;
```

```
    int seek = 0, index, min, distance;
```

```
    printf("Enter Number of Requests: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter Request Queue:\n");
```

```
    for (i = 0; i < n; i++) {
```

```
        scanf("%d", &req[i]);
```

```
    }
```

```
    printf("Enter Initial Head Position: ");
```

```
    scanf("%d", &head);
```

```
    while (count < n) {
```

```
        min = 9999;
```

```
        for (i = 0; i < n; i++) {
```

```
            if (!visited[i]) {
```

```
                distance = abs(req[i] - head);
```

```
                if (distance < min) {
```

```
                    min = distance;
```

```
                    index = i;
```

```
                }
```

```
            }
```

```
        }
```

```
        seek += min;
```

```
        head = req[index];
```

```
visited[index] = 1;
count++;
}
printf("Total Head Movement = %d\n", seek);
return 0; }
```

#### SHELL SCRIPT

```
GNU nano 8.7.1
#!/bin/bash

echo "SSTF Disk Scheduling Demonstration"

echo "Request Queue : 98 183 37 122 14 124 65 67"

echo "Initial Head Position : 53"

echo "Total Head Movement calculated using SSTF."
```

#### Output

```
(ganesh@LAPTOP-R3RDH9NG)~$ nano sstf.sh

(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x sstf.sh

(ganesh@LAPTOP-R3RDH9NG)~$ bash sstf.sh
SSTF Disk Scheduling Demonstration
Request Queue : 98 183 37 122 14 124 65 67
Initial Head Position : 53
Total Head Movement calculated using SSTF.
```